Sharp Influence and Adaptive Observation on Arbitrary Product Spaces

Quadratic matching, higher-degree backdoors,
cubic networks, and effective-rank recovery

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Abstract

Boundedness and independence impose dimension-free constraints on low-order interactions even when input alphabets are arbitrarily large or nonatomic. We prove that every real $|f| \le B$ of Hoeffding degree at most two on a product probability space satisfies

$$\sup_i \mathrm{I}_i(f) \ge \frac{\mathrm{Var}(f)^2}{C_2 B^2}, \quad C_2 = (10 + 2\sqrt{2})^2(1+\sqrt{3})^2 < 1229,$$

and the variance exponent two is optimal. The proof uses a coupling bound for disjoint interaction pairs, boundedness-derived section budgets, a greedy matching assignment, and independent Hilbert fourth moments. No Boolean-cube influence theorem is imported.

The same mechanism yields a *quadratic-backdoor principle*: if t coordinates intersect every interaction support of order at least three, then $\mathrm{Var}(f) \le \sqrt{C_2} B \sqrt{M} + tM$, where $M = \sup_i \mathrm{I}_i(f)$, irrespective of the original interaction degree. A stronger, sectional-backdoor hypothesis also gives a recovery algorithm: query the t backdoor coordinates and then use at most $12C_2 r B^2/\eta + 1$ adaptive whole-symbol queries to predict a Hilbert-valued target up to covariance-tail risk $\tau_r + \eta$. For degree-three functions the backdoor hypotheses coincide. An independent-block theorem allows arbitrarily many disjoint cubic networks, each having a bounded local cover, while preserving the variance-squared exponent and an alphabet-free scalar query rate. Rank-linear quadratic influence, spectral oracle inequalities, matching query-order lower bounds, and unbounded adaptive/nonadaptive separations complete the picture. We prove why an averaging backdoor alone does *not* yield a dimension-free observation budget at arbitrary degree. These are author-side mathematical arguments with exact finite-instance checks, not externally refereed or end-to-end Lean certified; the unrestricted cubic exponent-two problem remains open.

Keywords: quadratic backdoor; cubic support transversal; Hoeffding decomposition; Efron–Stein influence; bounded quadratic functions; arbitrary product laws; Hilbert-valued prediction; posterior energy; adaptive query complexity; covariance spectrum; quantum Bayesian gain.
Mathematics classification (provisional): 60E15, 68Q17, 60G15, 81P45.
Research status: self-contained analytic draft; proof and novelty review outstanding; no claim of kernel certification or external submission.

1 Introduction and the main phenomenon

A bounded observable may depend on thousands of independent inputs while changing very little when any one input is resampled. A quantitative influence theorem asks how small the largest coordinate influence can be relative to the total variance and the interaction order. If f is a

polynomial of degree two on fair binary signs, the Fourier representation makes the interaction structure apparent. The same mathematical question on a product of categorical, biased, or nonatomic probability spaces is more delicate: local contrast spaces can have arbitrarily large dimension, and the smallest atom mass can be arbitrarily close to zero.

This article develops a direct degree-two theory and a structural extension to covered higher-order interactions in that general setting. A four-corner comparison controls the total strength of *disjoint* pair interactions; a descending matching then distributes all pair interactions to endpoints whose accumulated costs remain bounded. Independence among the assigned neighboring coordinates makes a fourth-moment argument work in Hilbert space. Crucially, no fair-bit encoding, threshold certificate, or Boolean-cube influence theorem enters the proof.

The support structure provides a second, nontrivial bridge: an inexpensive set of backdoor coordinates can reduce high-order interactions to quadratic ones after those coordinates are revealed. A weaker hitting-set condition yields an influence theorem but not this observation theorem.

The same quadratic inequality has another consequence. If the output is a vector or Hilbert-space element and its covariance energy concentrates in a few principal directions, an adaptive observer need not read all input coordinates. The observer repeatedly chooses an influential remaining coordinate and predicts from its conditional mean. A degree-weighted posterior potential allows the resulting risk to telescope; a rank-sensitive version of the influence inequality produces a near-minimax query budget.

Relationship to previous work. Hoeffding decompositions and coordinate-resampling influences on general products are classical; product-space learning was studied by Blais–O'Donnell–Wimmer [4], while O'Donnell–Saks–Schramm–Servedio [3] established influential-coordinate inequalities for decision trees on arbitrary products, including non-Boolean outputs. The bounded polynomial influence question originated with Aaronson–Ambainis [5]; Turturean [11] recently proved an all-degree uniform-Boolean-cube theorem. Distributional variants for fixed Bernoulli bias and slices were analyzed by Girish–Mittal–Nehoran–Raz [7]. Those works provide context and deserve explicit credit; *none is an input to the proofs in this article.* The degree-two and covered-cubic results are more specialized but quantitatively stronger in the variance exponent for their classes, over a broader family of product input laws. Earlier completely bounded quadratic and Grothendieck methods provide additional, often sharper, results under different norms[10, 9]; a scope-matched priority search remains necessary.

Contributions and limits. Our results follow two independent structural paths:

$$\text{quadratic matching} \Longrightarrow \text{variance-squared influence} \Longrightarrow \text{rank-linear recovery},$$
$$\text{bounded interaction backdoor} \Longrightarrow \text{covered cubic influence and observation}.$$

The main results are existential over known distributions and known targets. For structured higher degree, the distinction between averaging away backdoor coordinates and actually observing them is indispensable. They do not assert an efficient algorithm for computing conditional expectations, influences, or optimal query decisions; nor do they extend to arbitrary correlated input distributions. The quantum consequences use two explicitly different access models, with their own assumptions.

2 Product-space notation and theorem statements

Let J be an arbitrary index set and $(\Omega_i, \mu_i)_{i\in J}$ a family of probability spaces. We use the usual product sigma algebra, or its completion, with joint law $\mu = \bigotimes_{i\in J} \mu_i$. The real Hilbert space $L^2(\mu)$ contains the orthogonal Hoeffding subspaces indexed by finite subsets $S \subseteq J$. On each coordinate, E_i denotes conditional integration against μ_i, and $D_i = \mathrm{Id} - E_i$.

Definition 2.1 (Interaction degree and influence)**.** A real or real-Hilbert-valued $F \in L^2(\mu; \mathcal{H})$ has Hoeffding interaction degree at most d if it lies in the closed linear span of the Hoeffding subspaces supported on finite S with $|S| \le d$. Its coordinate resampling influence and variance are

$$\mathrm{I}_i(F) = \|D_i F\|_{L^2}^2 = \tfrac{1}{2}\mathbb{E}\|F(X) - F(X^{(i)})\|_{\mathcal{H}}^2, \quad V(F) = \mathbb{E}\|F - \mathbb{E}F\|_{\mathcal{H}}^2. \tag{1}$$

Here $X^{(i)}$ is obtained from X by resampling the i-th coordinate independently. For a real F, $V(F) = \mathrm{Var}(F)$.

On finite products the unique Hoeffding expansion $F = \sum_S F_S$ satisfies

$$V(F) = \sum_{S\neq\varnothing} \|F_S\|_2^2, \qquad \mathrm{I}_i(F) = \sum_{S\ni i} \|F_S\|_2^2, \qquad V(F) \le \sum_i \mathrm{I}_i(F) \le dV(F). \tag{2}$$

The last inequality extends to arbitrary index sets by applying it on finite subsets. For $d = 2$, the nonconstant components consist of $h_i(X_i)$ and separately centered kernels $h_{ij}(X_i, X_j)$. We will write

$$D_2 = (10 + 2\sqrt{2})(1 + \sqrt{3}), \qquad C_2 = D_2^2 = 432 + 160\sqrt{2} + 216\sqrt{3} + 80\sqrt{6}. \tag{3}$$

Numerically $C_2 = 1228.356\ldots$. The exact expression governs every quantitative assertion in this paper.

Theorem 2.2 (Distribution-free quadratic influence)**.** *Let f be real, $|f| \le B$ almost surely and of Hoeffding degree at most two on any independent product probability space. If $B > 0$ and $V = \mathrm{Var}(f) > 0$, then the maximum coordinate influence is attained and*

$$\boxed{\max_{i\in J} \mathrm{I}_i(f) \ge \frac{V^2}{C_2 B^2}.} \tag{4}$$

The coefficient has no dependence on the index-set cardinality, coordinate biases, alphabet sizes, or minimum atom probabilities.

Proposition 2.3 (Sharpness of the variance exponent)**.** *For each integer $m \ge 1$, a bounded exact-degree-two function on $2m$ independent fair signs has $V = 1/m$ and $\max_i \mathrm{I}_i = 1/m^2$. Consequently the exponent two in (4) cannot be replaced uniformly by any smaller exponent.*

Definition 2.4 (Averaging transversal and sectional quadratic backdoor)**.** Let $\mathscr{S}_{\ge 3}(F)$ denote the sets S with $|S| \ge 3$ and nonzero Hoeffding component F_S. A finite set H is an *averaging transversal* if $S \cap H \neq \varnothing$ for every $S \in \mathscr{S}_{\ge 3}(F)$. It is a *sectional quadratic backdoor* if $|S \setminus H| \le 2$ for every nonzero Hoeffding support S (of any order). The latter condition is stronger in general, but *equivalent* to the former when F has degree at most three.

Theorem 2.5 (Higher-degree influence with a finite averaging transversal)**.** *Let f be real, $|f| \le B$, on an independent product of arbitrary probability spaces; no bound on its Hoeffding degree is required. Suppose an averaging transversal H has $|H| = t$. Put $V = \mathrm{Var}(f)$ and $M = \sup_i \mathrm{I}_i(f)$. Then*

$$V \le D_2 B\sqrt{M} + tM \le (D_2 + t)B\sqrt{M}, \qquad \boxed{M \ge \frac{V^2}{(D_2 + t)^2 B^2}.} \tag{5}$$

For a Hilbert output F with affine-range dimension at most r, $\|F\| \le B$, the analogous bound is

$$V(F) \le D_2 B\sqrt{3rM(F)} + tM(F) \le (D_2\sqrt{3r} + t)B\sqrt{M(F)}. \tag{6}$$

The $t = 0$ statements reduce to the quadratic influence theorems.

Theorem 2.6 (Sectional-backdoor observation at additive overhead)**.** *Let F be a known, Hilbert-valued bounded observable of n finite independent inputs, $\|F\| \le B$, with a sectional quadratic backdoor H of size t. For any $r \ge 1$ and $\eta > 0$ there is a deterministic adaptive whole-symbol observer with expected squared error at most $\tau_r(F) + \eta$, using at most*

$$\boxed{\min\{n,\ t + \lceil 12 C_2 r B^2/\eta \rceil\}} \tag{7}$$

coordinate reads. If $F = f$ is scalar, the factor 12 may be replaced by 4 and no output spectral tail is needed. In particular the statement holds for any bounded degree-three *target whenever H meets each nonzero cubic support, irrespective of alphabet sizes.*

Theorem 2.7 (Independent-block covered cubic networks)**.** *Let $f = c + \sum_{b=1}^{k} f_b(X_{J_b})$, with disjoint nonempty blocks J_b of independent coordinates, and assume $|f| \le B$ almost surely. Each block has Hoeffding degree at most three and possesses a cover $H_b \subset J_b$ of its nonzero cubic supports with $|H_b| \le t$. The number k and all coordinate alphabet sizes are arbitrary. Then*

$$\boxed{\max_i \mathrm{I}_i(f) \ge \frac{\mathrm{Var}(f)^2}{(D_2 + t)^2 B^2}.} \tag{8}$$

The same holds for countably many independent blocks under L^2 convergence and the boundedness hypothesis. For a finite product, there is a deterministic adaptive whole-symbol observer with risk at most $\eta > 0$ after at most

$$\boxed{\min\{n, \lceil 9(D_2 + t)^2 B^2/\eta \rceil\}} \tag{9}$$

queries. Neither bound depends on k or the leaf alphabet sizes.

Remark 2.8 (The two backdoor conditions are not interchangeable)**.** *At degree at most three, a transversal of all cubic supports is exactly a sectional quadratic backdoor. At higher degree an averaging transversal may leave arbitrarily high-degree sections after it is observed. Theorem 2.5 requires only the weak condition; Theorem 2.6 explicitly requires the strong one. A parity obstruction is proved in Proposition 4.4.*

Theorem 2.9 (Rank-linear Hilbert influence)**.** *Let G take values in a real Hilbert space and assume its essential range lies in an affine subspace of dimension at most r, with $r \ge 1$. If $\|G\| \le B$ almost*

surely and the input degree is at most two, then

$$\boxed{V(G)^2 \le 3C_2 r B^2 \max_i \mathrm{I}_i(G).} \tag{10}$$

For finite input index sets the maximum exists; for arbitrary index sets with $V(G) > 0$ it is attained by the same finite-superlevel argument as in Theorem 2.2.

Theorem 2.10 (Rank-sensitive adaptive observation)**.** *Let F be known, bounded by B in a real Hilbert space, of input degree at most two, on a product of $n < \infty$ independent coordinates. Let $\lambda_1 \ge \lambda_2 \ge \cdots \ge 0$ be its covariance eigenvalues and $\tau_r(F) = \sum_{j>r} \lambda_j$. For every $r \ge 1$ and $\eta > 0$ there exists a deterministic adaptive whole-symbol querying policy of depth at most*

$$N \le \min\left\{n, \left\lceil \frac{12C_2 r B^2}{\eta} \right\rceil\right\} \tag{11}$$

with a conditional-mean prediction $\widehat{F}$ satisfying $\mathbb{E}\|F - \widehat{F}\|^2 \le \tau_r(F) + \eta$. For each $r, k \ge 1$ there exists a degree-one $\|F\| \le 1$ example with $\tau_r = 0$ and target risk $\eta = 1/(2k)$ requiring at least $r/(4\eta)$ coordinate observations even for adaptive policies.

All four statements have proofs below using only independent-input conditioning and standard Hilbert-space inequalities. Theorem 2.10 is a query-count theorem, not a runtime guarantee.

3 Proof of the degree-two influence theorem

We first work with finite full-support product laws. Section 11 supplies the approximation argument for arbitrary product probability spaces. Write

$$f = \mathbb{E}f + \sum_i h_i(X_i) + \sum_{i<j} h_{ij}(X_i, X_j), \quad E_i h_i = 0, \quad E_i h_{ij} = E_j h_{ij} = 0. \tag{12}$$

For each i, put $H_i = L^2(\mu_i; \mathbb{R})$ and

$$U_i(\cdot; X_{-i}) = h_i(\cdot) + \sum_{j \ne i} h_{ij}(\cdot, X_j), \quad \mathrm{I}_i(f) = \|h_i\|_2^2 + \sum_{j \ne i} \|h_{ij}\|_2^2. \tag{13}$$

For a pair (i, j) set $b_{ij} = \max_{x_i, x_j} |h_{ij}(x_i, x_j)|$.

3.1 The disjoint-edge coupling budget

Lemma 3.1 (Matching mass)**.** *For any matching P in the graph of nonzero quadratic interactions, $\sum_{(i,j)\in\mathsf{P}} b_{ij} \le 4B$.*

Proof. Fix one edge (i, j) and a pair (a, b) realizing $|h_{ij}(a, b)| = b_{ij}$. Since h_{ij} is centered separately in both inputs, the average of $\Delta_{ij} = h_{ij}(a, b) - h_{ij}(a', b) - h_{ij}(a, b') + h_{ij}(a', b')$ over independent $a' \sim \mu_i$, $b' \sim \mu_j$ equals $h_{ij}(a, b)$. Choose a', b' so that $|\Delta_{ij}| \ge b_{ij}$. For the chosen edge compare the *even* two-point coupling, uniform on $(a, b), (a', b')$, to the *odd* coupling, uniform on $(a, b'), (a', b)$. They share the same one-coordinate marginals and their h_{ij} expectations differ by $\Delta_{ij}/2$. Independently

make this comparison on every edge of the matching, orienting each difference positively; give all unmatched coordinates the same law in both experiments. All additive terms and all interactions between distinct blocks have identical expectations because the marginals agree and blocks are independent. Hence the difference of the two expectations of f is $\frac{1}{2}\sum_{e\in\mathsf{P}}|\Delta_e|$. Both expectations lie in $[-B, B]$, so this difference is at most $2B$, proving the lemma. □

3.2 Two norm budgets from pointwise boundedness

Lemma 3.2 (Additive oscillation and coordinate-section budgets)**.** *For the decomposition* (12)*,*

$$\sum_i \|h_i\|_2 \le 2(1+\sqrt{2})B, \qquad \sum_i \mathbb{E}\|U_i(\cdot; X_{-i})\|_{H_i} \le 8B. \tag{14}$$

Proof. For the first inequality consider the product noise operator $T_t = \prod_i (E_i + tD_i)$, $0 \le t \le 1$. It is a positive Markov contraction, and $T_t f(x) = \mathbb{E}f + t\sum_i h_i(x_i) + t^2 \sum_{i<j} h_{ij}(x_i, x_j)$ lies in $[-B, B]$. For two inputs x, y, the difference $r(t) = T_t f(x) - T_t f(y) = ut + vt^2$ satisfies $r(0) = 0$, $|r(t)| \le 2B$, and $u = \sum_i [h_i(x_i) - h_i(y_i)]$. At $s = \sqrt{2} - 1$, exact interpolation gives $u = [r(s) - s^2 r(1)]/[s(1-s)]$, whence $|u| \le 2B(1+s^2)/[s(1-s)] = 4(1+\sqrt{2})B$. The oscillation of the additive sum is the sum of its coordinate oscillations on the Cartesian support. Since every centered scalar random variable in an interval of width w has standard deviation at most $w/2$, $\sum_i \|h_i\|_2 \le \frac{1}{2}\operatorname{osc}(\sum h_i) \le 2(1+\sqrt{2})B$.

For the second inequality, fix x and in each fiber choose a symbol maximizing $f(x_{-i}, \cdot) - f(x)$, with gain $a_i^+ \ge 0$, and one minimizing it, with loss $a_i^- \ge 0$. Independently switch each coordinate from its current symbol to its selected replacement with probability t. The resulting expected value $q_\pm(t)$ is a quadratic polynomial bounded by B for $0 \le t \le 1$, and $q'_+(0) = \sum_i a_i^+$, $q'_-(0) = -\sum_i a_i^-$. The identity $q'(0) = -3q(0) + 4q(1/2) - q(1)$ yields $\sum_i a_i^\pm \le 8B$. Thus the sum of fiber oscillations at every x is at most $16B$. Conditional on X_{-i}, the centered section U_i is the fiber value minus its mean, so its H_i-norm is at most half its range. Integrating and summing proves the second inequality. □

3.3 Greedy edge assignment and a fourth-moment inequality

Lemma 3.3 (Quadratic energy bound)**.** *Under* (12)*, with* $M = \max_i \mathrm{I}_i(f)$*,*

$$\sum_{i<j} \|h_{ij}\|_2^2 \le \left[8 + \sqrt{3}(8 + 2(1+\sqrt{2}))\right] B\sqrt{M}. \tag{15}$$

Proof. Process edges in nonincreasing order of b_{ij}, retaining an edge if disjoint from previously retained edges. The retained edges form a matching. Give both endpoints of a retained edge the cost $c_i = b_{ij}$ and give every unused vertex cost zero. Each original edge (i, j) can be assigned to an endpoint with $c_i \ge b_{ij}$: if not retained, it was blocked by an earlier retained edge of at least that weight. The matching lemma implies $\sum_i c_i \le 8B$.

Let S_i consist of the neighbors of edges assigned to i and put, in the Hilbert space H_i, $Y_{ij} = h_{ij}(\cdot, X_j)$, $Z_i = \sum_{j\in S_i} Y_{ij}$, and $s_i = \mathbb{E}\|Z_i\|_{H_i}^2 = \sum_{j\in S_i}\|h_{ij}\|_2^2 \le \mathrm{I}_i(f) \le M$. For a fixed i these H_i-valued random variables are independent and centered, with $\|Y_{ij}\|_{H_i} \le b_{ij} \le c_i$. Expanding the

fourth moment and using Cauchy–Schwarz,

$$\mathbb{E}\|Z_i\|_{H_i}^4 \le c_i^2 s_i + 3s_i^2. \tag{16}$$

Indeed the diagonal terms satisfy $\mathbb{E}\|Y_{ij}\|^4 \le c_i^2 \mathbb{E}\|Y_{ij}\|^2$, while the pairwise inner-product terms are at most products of their second moments. The interpolation $\|W\|_{L^2}^3 \le \|W\|_{L^1}\|W\|_{L^4}^2$ applied to $W = \|Z_i\|$ gives

$$s_i \le \sqrt{c_i^2 + 3s_i}\,\mathbb{E}\|Z_i\|_{H_i} \le (c_i + \sqrt{3M})\mathbb{E}\|Z_i\|_{H_i}. \tag{17}$$

Since Z_i is the conditional expectation of $U_i - h_i$ over the assigned neighbor coordinates, Jensen and the triangle inequality show $\mathbb{E}\|Z_i\| \le \mathbb{E}\|U_i\| + \|h_i\|_2$. Therefore Lemma 3.2 gives

$$\sum_i \mathbb{E}\|Z_i\| \le [8 + 2(1+\sqrt{2})]B, \qquad \mathbb{E}\|Z_i\| \le \sqrt{s_i} \le \sqrt{M}. \tag{18}$$

Every edge is assigned exactly once, so $\sum_{i<j}\|h_{ij}\|_2^2 = \sum_i s_i$. Sum (17), use $\sum c_i \le 8B$ and (18), and obtain (15). □

Proof of Theorem 2.2 for finite full-support products. The additive energy satisfies

$$\sum_i \|h_i\|_2^2 \le \sqrt{M}\sum_i \|h_i\|_2 \le 2(1+\sqrt{2})B\sqrt{M}.$$

By Hoeffding orthogonality and Lemma 3.3,

$$\begin{aligned} V &= \sum_i \|h_i\|_2^2 + \sum_{i<j}\|h_{ij}\|_2^2 \\ &\le [2(1+\sqrt{2}) + 8 + \sqrt{3}(8 + 2(1+\sqrt{2}))]B\sqrt{M} \\ &= D_2 B\sqrt{M}. \end{aligned}$$

Squaring proves $M \ge V^2/(C_2 B^2)$. □

Proof of Proposition 2.3. Let $Z_1, \dots, Z_{2m}$ be independent fair signs and $f_m = m^{-1}\sum_{k=1}^m Z_{2k-1}Z_{2k}$. The products are independent fair signs, so $|f_m| \le 1$ and $\mathrm{Var}(f_m) = 1/m$. Every coordinate occurs in one quadratic monomial of coefficient $1/m$, so its resampling influence is exactly $1/m^2$. A nonzero degree-two Fourier coefficient witnesses exact degree two. □

4 Higher-degree backdoors and disjoint cubic networks

The transfer mechanism is *remove a small set of coordinates and charge the removed variance to those coordinates*. This operation uses only Hilbert orthogonality and is different from encoding large alphabets into Boolean bits.

4.1 Averaging out higher-order interactions

Proof of Theorem 2.5. Write the Hoeffding expansion $f = \sum_S f_S$, with $f_\varnothing = \mathbb{E}f$, and let $g = \mathbb{E}[f \mid X_{J\setminus H}]$; that is, integrate over all coordinates in H. Every surviving support $S \subseteq J \setminus H$ has $|S| \le 2$,

since H meets every nonzero higher-order support. Thus g is quadratic and $|g| \le B$. For each $i \notin H$, $D_i g = \mathbb{E}[D_i f \mid X_{J\setminus H}]$, so conditional Jensen gives $\mathrm{I}_i(g) \le \mathrm{I}_i(f) \le M$. For $i \in H$ we have $\mathrm{I}_i(g) = 0$. Theorem 2.2 therefore yields $\mathrm{Var}(g) \le D_2 B\sqrt{M}$. The orthogonal variance loss is

$$\begin{aligned} V - \mathrm{Var}(g) &= \sum_{S:\, S\cap H\neq\varnothing} \|f_S\|_2^2 \\ &\le \sum_{i\in H}\sum_{S\ni i} \|f_S\|_2^2 = \sum_{i\in H} \mathrm{I}_i(f) \le tM. \end{aligned}$$

Hence $V \le D_2 B\sqrt{M} + tM$. Since $M \le V \le B^2$, $tM \le tB\sqrt{M}$, proving (5).

For Hilbert-valued F, replace the scalar quadratic inequality by Theorem 2.9; conditional expectation and orthogonality work unchanged in $L^2(\mu;\mathcal{H})$. Therefore $V(F) \le D_2 B\sqrt{3rM(F)} + tM(F)$, and $M(F) \le B^2$ gives (6). This proof works on arbitrary index sets by the same L^2 cylinder approximation used in Section 11; the support-cover assumption is preserved under finite conditional projections. No degree-three hypothesis was used for the influence conclusion. □

Corollary 4.1 (Exponent two for covered cubic targets)**.** *For a real $|f| \le B$ of Hoeffding degree at most three, if t coordinates meet every nonzero cubic support, then*

$$\max_i \mathrm{I}_i(f) \ge \mathrm{Var}(f)^2/[(D_2 + t)^2 B^2].$$

The same hypothesis on a Hilbert-valued F of affine output rank r gives $V(F)^2 \le (D_2\sqrt{3r} + t)^2 B^2 \max_i \mathrm{I}_i(F)$. The dependence on the variance has exponent two, strictly better than the exponent $5/2$ in the general-degree Boolean-cube theorem for $V/B^2 < 1$ within this restricted cubic class.

Remark 4.2 (Why the cover size matters)**.** *A small global transversal is not guaranteed for an arbitrary cubic interaction hypergraph. Theorem 2.5 must not be quoted with a constant independent of t unless t is known to be bounded. In particular, no statement about unrestricted cubic influence follows from this calculation.*

4.2 Observing the backdoor rather than averaging it

Proof of Theorem 2.6. The sectional hypothesis states that every nonzero component F_S contains at most two coordinates outside H. Reveal all t coordinates in H first. For almost every realized tuple $X_H = x_H$, the remaining-input section F_{x_H} is of Hoeffding degree at most two: restriction of an original component F_S depends only on $S \setminus H$ outside H, and $|S \setminus H| \le 2$. By Fubini, $\|F_{x_H}\| \le B$ almost surely on the residual independent product, and its output still lies in the same Hilbert space.

Fix the *unconditional* covariance rank-r projection P_r of F and set $G = P_r F$. It remains bounded by B and sectional quadratic after revealing H. On every such posterior, apply Theorem 2.10 to G_{x_H} for a further $q = \lceil 12 C_2 r B^2/\eta \rceil$ adaptive reads, unless all coordinates are exhausted. Conditional expected residual risk of G is at most η (and is zero if the full input is read). The fixed orthogonal projection $F - G$ has expected conditional variance at most its unconditional variance $\tau_r(F)$ by the law of total variance; cross terms vanish in orthogonal output subspaces. Sum the risks. For real scalar f, use Corollary 6.2, replacing the coefficient 12 by 4. The strategy only needs the known backdoor H and the quadratic posterior influence oracle; its computational runtime is not bounded. For degree-three input, the two backdoor notions in Definition 2.4 coincide. □

Corollary 4.3 (A spectral oracle inequality after a backdoor)**.** *Under the preceding hypotheses, if $q > t$ observations are allowed, then the optimal expected squared error obeys*

$$\mathcal{R}_q(F) \le \inf_{r\ge 1}\left\{\tau_r(F) + \frac{12C_2 r B^2}{q-t}\right\}, \tag{19}$$

with zero risk for $q \ge n$. For scalar outputs replace 12 *by* 4*. In particular, a bounded-size quadratic backdoor adds only an* additive *observation cost, even when the original input interaction degree is larger than three.*

Proof. Choose $\eta = 12C_2 r B^2/(q-t)$ in Theorem 2.6; the extra quadratic observation budget is at most $q - t$. Take the infimum over r. The scalar variant is identical. □

Proposition 4.4 (Why averaging alone does not imply observation)**.** *For every $m \ge 3$ there exists a $\{-1, 1\}$-valued input function f_m with an averaging transversal of size one such that any adaptive observer that reads fewer than $m + 1$ whole coordinates has squared conditional-mean error exactly one. This function has* no *sectional quadratic backdoor of size one for $m \ge 4$.*

Proof. Let $X_0, \dots, X_m$ be independent fair signs and $f_m(X) = \prod_{j=0}^{m} X_j$. Its only nonconstant Hoeffding support is $S = \{0, \dots, m\}$, which meets $H = \{0\}$; hence H is an averaging transversal. However, after any proper subset of coordinates is observed, at least one independent fair sign remains. The conditional mean of f_m is exactly zero and its squared conditional error is one. Thus $m + 1$ observations are necessary even for an adaptive algorithm. With $H = \{0\}$, $|S \setminus H| = m > 2$, so H is not sectional quadratic (and no other singleton is). □

4.3 Independent cubic networks with no bounded global cover

Proof of Theorem 2.7. First consider finitely many blocks. For each independent block observable f_b, let R_b be the width of its essential range. Independence and the pointwise bound $|f| \le B$ force

$$\sum_b R_b \le 2B. \tag{20}$$

Indeed, the essential supremum of a sum of independent real random variables is the sum of their essential suprema, and similarly for the infima. Apply these equalities to the independent block outputs; subtracting gives (20). In particular each R_b is finite. Translate each f_b by its range midpoint; the shifted function g_b has $|g_b| \le B_b = R_b/2$. Translation changes neither its Hoeffding degree nor its variance or coordinate influences. Corollary 4.1 applies block by block. With $M = \max_i \mathrm{I}_i(f)$, each block influence is at most M, so

$$\mathrm{Var}(f) = \sum_b \mathrm{Var}(f_b) \le (D_2 + t)\sqrt{M} \sum_b B_b \le (D_2 + t) B \sqrt{M}.$$

This proves (8). For countably many independent blocks, apply the argument to finite partial sums and use the monotone variance convergence and the finite-prefix range bound inherited from the full bounded sum; the asserted L^2 convergence supplies the limiting identity.

For adaptive observation, after any transcript, the remaining unobserved coordinates are still independent, and within each block there are at most three active variables per support, with every

surviving cubic support meeting the unobserved part of the original H_b. The conditional section remains bounded by B and retains an independent-block representation. Hence the scalar influence bound with constant $(D_2 + t)$ applies at *every* posterior. For the current posterior let v be its variance, p its degree-weighted Hoeffding energy and m its maximal resampling influence. Then $v \le p \le 3v$, and the already-proved influence bound implies

$$m \ge \frac{v^2}{(D_2+t)^2 B^2} \ge \frac{p^2}{9(D_2+t)^2 B^2}.$$

The posterior energy decrement identity of Lemma 6.1 holds verbatim for degree three: revealing a coordinate reduces the expected degree weight of each support that contains it by exactly one. The greedy policy therefore obeys $\mathbb{E}p_{q+1} \le \mathbb{E}p_q - (\mathbb{E}p_q)^2/[9(D_2+t)^2B^2]$. Reciprocal telescoping gives $\mathbb{E}p_q \le 9(D_2+t)^2B^2/q$. Since the conditional mean-square risk is at most $\mathbb{E}p_q$, (9) follows, capped by n. □

Proposition 4.5 (Sharp variance exponent in genuine cubic networks)**.** *For each $m \ge 2$ let A, B be independent uniform variables on $[m]$, and let $(Z_{ab})_{a,b\in[m]}$ be independent fair signs. Set $h(A,B,Z) = Z_{A,B}$. Then h has exact Hoeffding degree three, $|h| = 1$, $\mathrm{Var}(h) = 1$, and*

$$\mathrm{I}_A(h) = \mathrm{I}_B(h) = 1 - 1/m, \qquad \mathrm{I}_{Z_{ab}}(h) = 1/m^2. \tag{21}$$

For k independent such blocks, the bounded average $f_k = k^{-1}\sum_{b=1}^{k} h_b$ has variance $1/k$ and maximum influence $(1-1/m)/k^2$. Thus the exponent two in Theorem 2.7 is optimal even on genuine degree-three covered cubic networks. Three adaptive queries recover h exactly, whereas any fixed nonadaptive strategy attaining squared error at most $\eta < 1$ must query at least $(1-\eta)m^2$ leaves, even if both addresses are subsequently revealed at no cost.

Proof. The exact Hoeffding expansion follows from $\mathbf{1}_{\{A=a\}} = m^{-1} + (\mathbf{1}_{\{A=a\}} - m^{-1})$ and the analogous expansion for B:

$$h = \sum_{a,b} \left(m^{-1} + \left[\mathbf{1}_{\{A=a\}} - m^{-1}\right]\right)\left(m^{-1} + \left[\mathbf{1}_{\{B=b\}} - m^{-1}\right]\right) Z_{ab}.$$

The product of both centered address contrasts with Z_{ab} has nonzero third-order energy. Conditional on all other coordinates, resampling address A changes the selected independent sign unless the fresh address equals the previous one, so $\mathrm{I}_A = 1 - 1/m$; symmetry gives I_B. Only the event $(A,B) = (a,b)$ exposes leaf Z_{ab}, giving $\mathrm{I}_{Z_{ab}} = 1/m^2$. Independence of the k copies gives $V(f_k) = 1/k$ and the displayed maximal influence after the $1/k$ scaling. For nonadaptive selection of a set L of q leaves, even if both addresses are supplied afterwards, the conditional predictor is the chosen leaf when $(A,B) \in L$ and zero otherwise; the risk is $1 - q/m^2$. Three adaptive reads, A, B and then $Z_{A,B}$, recover h exactly. This is an elementary illustrative construction; no originality is asserted for the address trick. □

Remark 4.6 (The genuine cubic open problem)**.** *The independent-block theorem requires disjoint input blocks, not merely disjoint degree-three hyperedges. For arbitrary interlocking cubic support hypergraphs there may be no bounded cover, and the neighbor increments used in the quadratic fourth-moment proof may share random coordinates and cease to be independent. Neither theorem*

establishes a universal cubic variance-squared bound. The large-atom/link-overlap lemma *remains a plausible route but is not proved in this manuscript.*

5 Proof of the rank-linear degree-two theorem

The scalar proof is stable under Hilbert-valued interaction increments because it uses independent fourth moments rather than independent scalar coordinates of the output.

Lemma 5.1 (Finite-sign Hilbert compression)**.** *For $w_1, \dots, w_r$ in any real Hilbert space $\mathcal{K}$ and independent fair signs ε_j,*

$$\mathbb{E}_\varepsilon \|\sum_{j=1}^{r} \varepsilon_j w_j\|_{\mathcal{K}} \geq \frac{1}{\sqrt{3}} \left(\sum_{j=1}^{r} \|w_j\|_{\mathcal{K}}^2 \right)^{1/2}. \tag{22}$$

Proof. Let $W = \sum_j \varepsilon_j w_j$ and $s = \sum_j \|w_j\|^2$. Independence gives $\mathbb{E}\|W\|^2 = s$ and

$$\begin{aligned} \mathbb{E}\|W\|^4 = \sum_j \|w_j\|^4 + 2 \sum_{j<k} \|w_j\|^2 \|w_k\|^2 \\ + 4 \sum_{j<k} \langle w_j, w_k \rangle^2 \leq 3s^2. \end{aligned}$$

Interpolation yields $s^{3/2} \leq (\mathbb{E}\|W\|)(\mathbb{E}\|W\|^4)^{1/2} \leq \sqrt{3}\, s \mathbb{E}\|W\|$. Divide by s, with $s = 0$ trivial. □

Proof of Theorem 2.9 on finite full-support products. If the essential range lies in an affine r-plane, orthogonally project G onto the associated direction subspace. Such a projection preserves its variance and all influences while not increasing its pointwise norm. Thus identify the effective linear output subspace with $\mathbb{R}^r$ and decompose G into vector-valued h_i, h_{ij}. For every sign vector $\varepsilon \in \{-1, 1\}^r$ the scalar $g_\varepsilon = \langle \varepsilon, G \rangle / \sqrt{r}$ satisfies $|g_\varepsilon| \leq B$. Apply Lemmas 3.1 and 3.2 to every such scalar function and average in ε.

For the matching bound, fix one point attaining $b_{ij} = \max \|h_{ij}\|$ on each edge of a prescribed matching before choosing ε. Lemma 5.1, first in $\mathbb{R}^r$, gives

$$\sum_{(i,j) \in \mathsf{P}} b_{ij} \leq 4B\sqrt{3r}. \tag{23}$$

For the additive and coordinate-section budgets apply the same sign lemma in the spaces $L^2(\mu_i)$ (to the r scalar components of h_i) and $L^2(\mu_i)$ (to each fiber of U_i), then integrate. The result is

$$\sum_i \|h_i\|_2 \leq 2(1 + \sqrt{2})B\sqrt{3r}, \qquad \sum_i \mathbb{E}\|U_i(\cdot; X_{-i})\|_{L^2(\mu_i; \mathbb{R}^r)} \leq 8B\sqrt{3r}. \tag{24}$$

Run the identical descending-matching assignment with vector-valued b_{ij}. It yields $\sum_i c_i \leq 8B\sqrt{3r}$. The assigned increments $Y_{ij} = h_{ij}(\cdot, X_j)$ are independent centered random elements of $L^2(\mu_i; \mathbb{R}^r)$, with $\|Y_{ij}\| \leq c_i$. Thus the exact same fourth-moment and interpolation argument (16)–(17) applies, and

$$\sum_{i<j} \|h_{ij}\|_2^2 \leq \left[8 + \sqrt{3}(8 + 2(1 + \sqrt{2})) \right] B\sqrt{3rM(G)}.$$

The additive energy is at most $2(1+\sqrt{2})B\sqrt{3rM(G)}$. Orthogonality gives $V(G) \le D_2 B\sqrt{3rM(G)}$, hence (10). □

Remark 5.2 (Why the rank factor is structurally meaningful)**.** *The argument does not assert that the individual components of G are independent. The $\sqrt{r}$ loss comes only from compressing bounded Hilbert outputs to bounded random scalar projections, while the assigned neighbor increments are independent because their* input coordinates *are independent. The lower bound in Section 6 shows that the induced linear r dependence in observation depth cannot generally be eliminated, although the constant is not claimed optimal.*

6 Proof of rank-optimal adaptive quadratic observation

The key fact for adaptive observation is not merely that each posterior has an influential coordinate. It is that a *degree-weighted residual energy* drops by exactly that coordinate's influence in conditional expectation.

Lemma 6.1 (Actual degree-weighted posterior drift)**.** *Let G be Hilbert-valued of input degree at most two on a finite product. At a transcript revealing coordinates Q, decompose the remaining-input section into orthogonal Hoeffding components G_S and write*

$$v_Q = \sum_{S\neq\varnothing} \|G_S\|_2^2, \qquad p_Q = \sum_{S\neq\varnothing} |S|\|G_S\|_2^2 = \sum_{i\notin Q} \mathrm{I}_i(G \mid Q).$$

Then $v_Q \le p_Q \le 2v_Q$. If an unobserved coordinate i is revealed, conditional on the current transcript,

$$p_Q - \mathbb{E}[p_{Q\cup\{i\}} \mid \textit{current transcript}] = \mathrm{I}_i(G \mid Q). \tag{25}$$

Proof. For each nonempty residual support S not containing i, the contribution G_S persists. A term $G_{S\cup\{i\}}$ that contains i becomes, after observation of X_i, a random coefficient on S, decreasing its degree weight by one. Terms now sharing a residual support S have cross-inner-product expectation zero over the newly revealed X_i, because $E_i G_{S\cup\{i\}} = 0$. Hence the expected new weighted energy is $\sum_{S:\, i\notin S} |S|\|G_S\|_2^2 + \sum_{S:\, i\in S}(|S|-1)\|G_S\|_2^2$, where the term with $S=\{i\}$ has new weight zero. Subtract from the old energy to get $\sum_{S\ni i}\|G_S\|_2^2 = \mathrm{I}_i(G \mid Q)$. The sandwich follows because each positive-degree term has degree one or two. □

Proof of the upper bound in Theorem 2.10. Let P_r project onto r leading covariance eigenvectors of F and set $G = P_r F$, so $\|G\| \le B$, degree at most two, and effective output dimension at most r. At every possible transcript the unrevealed coordinates retain their original independent laws and the section of G is still bounded and quadratic. Theorem 2.9 therefore gives, with $K_r = 3C_2 r B^2$,

$$m_Q := \max_{i\notin Q} \mathrm{I}_i(G \mid Q) \ge \frac{v_Q^2}{K_r} \ge \frac{p_Q^2}{4K_r}.$$

Choose at each transcript a coordinate attaining m_Q, breaking ties in a fixed way. Freeze branches

after all coordinates are queried. Let $P_t = \mathbb{E}[p_Q]$ at observation depth t. By Lemma 6.1 and Jensen,

$$P_t - P_{t+1} = \mathbb{E}[m_Q] \geq \frac{\mathbb{E}[p_Q^2]}{4K_r} \geq \frac{P_t^2}{4K_r}. \tag{26}$$

Until $P_t = 0$, reciprocal telescoping gives $P_t \leq 4K_r/t$ for $t \geq 1$. The conditional-mean predictor of G has squared risk $\mathbb{E}v_Q \leq P_t$. Thus $t \geq 4K_r/\eta = 12C_2rB^2/\eta$ suffices. Reading all n coordinates gives zero error for G, producing the cap in (11).

The component $F - G$ is pointwise orthogonal to G in the fixed output decomposition. Its posterior conditional variance is at most its unconditional variance $\tau_r(F)$ by the law of total variance, and the cross term in conditional squared norm vanishes by the output-space orthogonality. The total risk is at most $\tau_r(F) + \eta$. □

Proof of the lower bound in Theorem 2.10. Let Z_{jt} $(1 \leq j \leq r,\ 1 \leq t \leq k)$ be independent fair signs and define

$$F(Z) = \frac{1}{k\sqrt{r}} \left(\sum_{t=1}^{k} Z_{1t}, \ldots, \sum_{t=1}^{k} Z_{rt} \right) \in \mathbb{R}^r. \tag{27}$$

Every input satisfies $\|F\| \leq 1$, the function has degree one, its covariance rank is r and its total variance is $1/k$. Each observed sign reduces the conditional squared variance by exactly $1/(rk^2)$, regardless of other observations or adaptivity. To reduce the risk to $1/(2k)$ requires at least $rk/2 = r/(4\eta)$ observations. □

Corollary 6.2 (Sharper scalar adaptive observation)**.** *For a real scalar degree-two target $|f| \leq B$ on a finite product, and every $\eta > 0$, there is a deterministic adaptive policy with expected squared error at most η after at most $\min\{n, \lceil 4C_2B^2/\eta \rceil\}$ whole-coordinate observations.*

Proof. Use the posterior potential p_Q of Lemma 6.1. The scalar Theorem 2.2 has $K = C_2B^2$, rather than the Hilbert rank bound $K_r = 3C_2rB^2$. Therefore at each transcript $m_Q \geq v_Q^2/K \geq p_Q^2/(4K)$. The exact drift and reciprocal telescoping from Theorem 2.10 give $\mathbb{E}p_Q \leq 4K/t$, and $\mathbb{E}v_Q \leq \mathbb{E}p_Q$. □

7 Spectrum-adaptive prediction and query allocation

The fixed-rank theorem is strongest when the output genuinely lives in a low-dimensional space, but the same calculation automatically adapts to approximately low-rank outputs. This is relevant when the output is a function, a probability vector, or a vector of correlated target statistics.

Theorem 7.1 (Covariance-tail oracle inequality)**.** *Under the hypotheses of Theorem 2.10, let $\mathcal{R}_q(F)$ denote the least expected squared error among deterministic policies using at most $q \geq 1$ coordinate observations. Then*

$$\boxed{\mathcal{R}_q(F) \leq \inf_{r \geq 1} \left\{ \tau_r(F) + \frac{12C_2rB^2}{q} \right\}.} \tag{28}$$

If $q \geq n$ the optimal risk is exactly zero. If $\operatorname{rank}(\operatorname{Cov} F) \leq r_0$, then $\mathcal{R}_q(F) \leq 12C_2r_0B^2/q$.

Proof. Fix an integer $r \geq 1$. Choose $\eta = 12C_2rB^2/q$ in Theorem 2.10. The resulting query cap is $\min\{n, \lceil q \rceil\}$ and the risk is at most $\tau_r + \eta$. Take the infimum over r. Reading all inputs gives the separate assertion when $q \geq n$. □

Corollary 7.2 (Polynomial spectral tails)**.** *Suppose $\tau_r(F) \le Ar^{-\alpha}$ for constants $A > 0$, $\alpha > 0$. For any $q \ge 1$, with $H = 12C_2B^2/q$, one has*

$$\mathcal{R}_q(F) \le \inf_{r\in\mathbb{N},\, r\ge 1}\{Ar^{-\alpha} + Hr\} \lesssim_\alpha A^{1/(\alpha+1)}H^{\alpha/(\alpha+1)} + H. \tag{29}$$

The implicit constant is explicit, depending only on α.

Proof. The real optimizer is $x_* = (\alpha A/H)^{1/(\alpha+1)}$. If $x_* \ge 1$, choose $r = \lceil x_* \rceil \le x_* + 1$. Then $Ar^{-\alpha} \le Ax_*^{-\alpha}$ and $Hr \le Hx_* + H$, yielding the stated order with coefficient $\alpha^{-\alpha/(\alpha+1)} + \alpha^{1/(\alpha+1)}$ before the additive H. When $x_* < 1$ choose $r = 1$ and absorb the result in the displayed right side. □

Corollary 7.3 (Exponential spectral tails)**.** *If $\tau_r(F) \le Ae^{-\beta r}$ for $\beta > 0$, then $\mathcal{R}_q(F) \le O_{A,\beta,B,C_2}((1+\log q)/q)$ for $q \ge 2$.*

Proof. Choose $r = \max\{1, \lceil \beta^{-1}\log(q)\rceil\}$ in (28); the first term is at most A/q and the second is $O((1+\log q)/q)$. □

What effective rank means here. The covariance spectrum is taken in the output Hilbert space, not the input space. If $F = (F_1, \dots, F_s)$ is a collection of observables whose outputs are almost collinear as random vectors, a large ambient s need not lead to a large query budget. Conversely, a small input interaction degree does not imply a small covariance rank: the lower-bound example in Theorem 2.10 has degree one and rank r as large as desired.

Policy description. For a chosen r, compute a rank-r covariance projection P_r and set $G = P_rF$. At a transcript $h = (Q, x_Q)$, form the posterior functions of G on the unrevealed independent coordinates, calculate $\mathrm{I}_i(G \mid h)$ for each $i \notin Q$, choose an index attaining the maximum, and reveal the entire symbol X_i. At termination, report $\mathbb{E}[F \mid h]$ (or $\mathbb{E}[G \mid h]$ plus the posterior mean of the orthogonal residual). This is a deterministic rule after fixed tie-breaking; its definition may demand expensive integration and optimization. It is not an efficiently executable sampler without additional structure in F or the source.

8 An unbounded adaptive–nonadaptive separation

The preceding theorems concern adaptive selection of whole coordinates. They do not automatically imply a small predetermined observation set, even when every coordinate has a finite alphabet and the target has interaction degree exactly two. The following elementary example isolates the obstruction.

Theorem 8.1 (Quadratic addressing separation)**.** *For every integer $m \ge 3$ there is a bounded, mean-zero, degree-two observable f of variance one on $m+1$ independent finite-alphabet coordinates such that two adaptive whole-symbol queries determine f exactly, whereas every nonadaptive fixed-set observation policy achieving expected squared error strictly less than $1/2$ queries more than $m/2$ coordinates.*

Proof. Let A be uniform on $[m]$, let $Z_1, \dots, Z_m$ be independent fair signs, independent of A, and define $f(A, Z) = Z_A$. Its expansion

$$f(A,Z) = \sum_{j=1}^{m} \mathbf{1}_{\{A=j\}} Z_j = \frac{1}{m} \sum_{j=1}^{m} Z_j + \sum_{j=1}^{m} \left(\mathbf{1}_{\{A=j\}} - \frac{1}{m} \right) Z_j \tag{30}$$

contains only order-one and order-two Hoeffding terms. It has mean zero, $|f| = 1$, and variance one. Querying A first identifies the required second coordinate Z_A, so two adaptive queries recover f with zero error.

For a fixed observation set, suppose first that it contains A and a set $S \subseteq [m]$ of s sign coordinates. The conditional predictor is $Z_A \mathbf{1}_{\{A \in S\}}$; the conditional mean-square risk averaged over A is $1 - s/m$. To obtain risk below $1/2$ requires $s > m/2$. If the observation set does not contain A, its conditional predictor is $m^{-1} \sum_{j \in S} Z_j$. Its squared L^2 norm is s/m^2, hence its Bayes risk is $1 - s/m^2 \geq 1 - 1/m > 1/2$. Randomizing a fixed query set before seeing any input is a convex combination of such risks and cannot improve this threshold. The same conclusion holds for arbitrary randomized prediction given a fixed set, because the conditional expectation already minimizes squared loss. □

Remark 8.2 (Not a quantum separation)**.** *The addressing example is classical. It distinguishes adaptive from fixed coordinate selection in a product prior; it does not establish a quantum advantage or a gap between collective and local quantum measurements. Its role is to delimit the observable notion to which the query upper bound applies. The construction is standard and no originality is claimed for it.*

9 A predetermined-set result above the quadratic interaction floor

A limited nonadaptive companion is possible when the target error is allowed to exceed the total pair-interaction energy.

Proposition 9.1 (Fixed-set observation above the quadratic floor)**.** *Let $f = \mathbb{E}f + \sum_i h_i + \sum_{i<j} h_{ij}$ be real, bounded by B, and of degree at most two on a finite product. Put $\delta_2 = \sum_{i<j} \|h_{ij}\|_2^2$. For each $\eta > \delta_2$ there exists a fixed set of at most*

$$\min \left\{ n, \left\lceil \frac{(3 + 2\sqrt{2})B^2}{\eta - \delta_2} \right\rceil \right\} \tag{31}$$

coordinates whose conditional-mean predictor has risk at most η.

Proof. Set $b_i = \|h_i\|_2$ and $L = \sum_i b_i$. Lemma 3.2 yields $L \leq 2(1 + \sqrt{2})B$. Read the q largest b_i. If $t = b_{q+1}$ then

$$\sum_{i>q} b_i^2 \leq t \sum_{i>q} b_i \leq t(L - qt) \leq \frac{L^2}{4q} \leq \frac{(3 + 2\sqrt{2})B^2}{q}.$$

All unrevealed pair energy is at most δ_2, and the risk is at most the sum of these two terms. Choose the displayed q, or read all coordinates if $q \geq n$. □

A structural obstruction. For the addressing function in (30), pair interactions carry most of its variance. Below the pair-interaction error floor, additive-importance rankings cannot guarantee a small fixed set. The adaptive policy instead uses the answer to one coordinate query to identify the next relevant interaction.

10 Optional quantum applications: distinct observation models

We give two distinct applications. The first queries the classical input symbols of a quantum circuit's output probability. The second actually measures physical encoded quantum registers. These observation contracts cannot be interchanged.

10.1 Predicting one-query quantum probability vectors

Let $P(X) = (P_1(X), \dots, P_s(X))$ be the output probability vector of a quantum algorithm that makes one coherent whole-symbol query to a finite-alphabet oracle. Under the standard oracle polynomial method, each final amplitude is a sum of terms depending on at most one classical coordinate, and therefore each outcome probability has Hoeffding interaction degree at most two [6].

Corollary 10.1 (One-query output probability prediction)**.** *For any known product distribution on the classical input symbols, any $\eta > 0$, and such a quantum algorithm with s labeled outcomes, there is a deterministic adaptive classical policy querying at most*

$$\min\left\{n, \left\lceil \frac{12C_2 s}{\eta} \right\rceil\right\}$$

whole input symbols and returning a probability vector $\widehat{P}$ with $\mathbb{E}\|P(X) - \widehat{P}(X)\|_2^2 \le \eta$.

Proof. Each P_j has interaction degree at most two and $\|P(x)\|_2 \le \|P(x)\|_1 = 1$. Apply Theorem 2.10 with $B = 1$, $r = s$, and $\tau_s = 0$. The conditional expectation of a probability vector remains in the probability simplex. □

Corollary 10.2 (Scalar one-query acceptance probability)**.** *If $g(X) \in [0,1]$ is a one-query acceptance probability, then for $0 < \epsilon, \delta < 1$ there is a deterministic classical $\widehat{g}$ with*

$$\Pr\{|g(X) - \widehat{g}(X)| > \epsilon\} \le \delta, \qquad N \le \min\left\{n, \left\lceil \frac{C_2}{\delta\epsilon^2} \right\rceil\right\}. \tag{32}$$

Proof. Apply the sharper scalar Corollary 6.2 to $f = 2g - 1$ with $B = 1$ and target $4\delta\epsilon^2$. Rescale the resulting posterior mean to $\widehat{g}$ and use Markov's inequality. □

Oracle-scope warning. The prediction policy is not a procedure for making optimal physical measurements of unknown quantum states. It presumes a known classical oracle input sampled from a known independent prior and access to the entire value of a selected input symbol, as opposed to a noisy quantum encoding of that symbol.

10.2 Quantum Bayesian terminal gain under local encodings

Let $X_1, \dots, X_n$ be finite independent classical variables with arbitrary positive-probability local laws. At site i, the symbol x_i is encoded into a density operator ρ_{i,x_i} on a finite-dimensional quantum register. The whole conditional state is $\rho_x = \bigotimes_i \rho_{i,x_i}$, and the faithful average on its support is $b_i = \mathbb{E}\rho_{i,X_i}$. For a real target f, let $G_t^*(Q)$ denote the optimal reduction in squared-error Bayes risk from a collective POVM on registers Q after each encoder is mixed with its mean state:

$$\rho_{i,x}(t) = t\rho_{i,x} + (1-t)b_i, \qquad 0 \le t \le 1.$$

The target and prior are assumed known to the measurement designer. Define $J_b(W) = (bW + Wb)/2$ and the local contrast map on centered classical functions

$$T_i g = \mathbb{E}[g(X_i)\rho_{i,X_i}], \qquad K_i = T_i^* J_{b_i}^{-1} T_i.$$

Assume $K_i \succeq aI$ on the *full centered local contrast space* at every coordinate on which f depends, for some $a > 0$. In particular, $0 < a \le 1$. This excludes a physically invisible contrast direction, even if other directions are visible.

Proposition 10.3 (Conditional quantum visibility transfer)**.** *For a bounded degree-two real target f with $V = \mathrm{Var}(f) > 0$, the terminal marginal value $\Delta_i(t) = G_t^*([n]) - G_t^*([n] \setminus \{i\})$ satisfies*

$$\boxed{\max_i \Delta_i(t) \;\ge\; \frac{t^4 a^2 V^2}{2C_2 B^2}.} \tag{33}$$

For binary source alphabets the factor 2 in the denominator may be omitted. The result concerns a register's marginal value when other registers are available*, not its one-register-from-scratch value.*

Proof. For product-prepared sources, the centered signal decomposes into orthogonal quantum ANOVA support components. The Personick–Sylvester operator for scalar quadratic loss identifies nonnegative optimal gains γ_S with

$$G_t^*(Q) = \sum_{\varnothing \ne S \subseteq Q} t^{2|S|}\gamma_S, \quad \Delta_i(t) = t^2\gamma_{\{i\}} + t^4 \sum_{j \ne i} \gamma_{\{i,j\}}.$$

The support decomposition follows by extending each local Sylvester solution by an identity on unused product-average registers; cross terms vanish by Hoeffding orthogonality. The one-site quadratic form equals $\langle f_i, K_i f_i\rangle$ and hence $\gamma_{\{i\}} \ge a\|f_i\|_2^2$. For two sites, diagonalize b_i, b_j and compare the entrywise inverse Jordan multipliers. For positive eigenvalues p, q, u, v,

$$\frac{2}{pu + qv} \;\ge\; \frac{1}{2}\frac{2}{p+q}\frac{2}{u+v},$$

because $(p+q)(u+v) \ge pu + qv$. Thus $J_{b_i \otimes b_j}^{-1} \succeq \frac{1}{2}(J_{b_i}^{-1} \otimes J_{b_j}^{-1})$, so $\gamma_{\{i,j\}} \ge \frac{1}{2}a^2\|f_{ij}\|_2^2$. As $a \le 1$ and $t^2 \ge t^4$,

$$\Delta_i(t) \ge \frac{1}{2}t^4 a^2(\|f_i\|_2^2 + \sum_{j \ne i}\|f_{ij}\|_2^2) = \frac{1}{2}t^4 a^2\, \mathrm{I}_i(f).$$

Apply Theorem 2.2. If each nonconstant classical contrast space is one-dimensional (binary sources),

the pair signal is a simple Hermitian tensor, for which the independent orientation-averaging inequality in Appendix A improves the comparison factor 1/2 to 1. This proves the stated binary refinement. □

Remark 10.4 (Do not conflate this with the trine conjecture)**.** *The visibility-only qubit extremality conjecture in Wright's companion note predicts a sharper transfer constant at fixed visibility. Its unrestricted qubit inequality has not been proved. Neither its conditional improvement nor a numerical qutrit counterexample is included as a theorem here. Similarly, the quantum proposition is analytic and not end-to-end Lean-certified.*

11 Finite-to-general transfer and independent proof scope

The matching argument was proved for finite products with finite alphabet support. The following approximation supplies the otherwise missing step for general products; it is particularly useful for non-atomic coordinates and uncountable index sets.

Lemma 11.1 (Bounded finite-partition approximation)**.** *Let J be any index set and $F \in L^2(\mu; \mathcal{H})$ be essentially bounded and of Hoeffding degree at most d, where $\mathcal{H}$ is a fixed finite dimensional real Hilbert space. There is a directed net of finite-product conditional expectations $P_\alpha F$ such that each is bounded by B, retains degree at most d, and converges to F in L^2. For each coordinate i, $\mathrm{I}_i(P_\alpha F) \le \mathrm{I}_i(F)$; moreover*

$$\sum_{i \in S} \mathrm{I}_i(F) \le d V(F)$$

for every finite $S \subseteq J$.

Proof. Let $\mathcal{A}_\alpha$ be the sigma algebra generated by a finite subset of coordinates together with finite measurable partitions of each chosen coordinate. Order these algebras by refinement. Finite rectangle indicators span a dense subspace of the product L^2; this is a monotone-class consequence of the fact that finite cylinders generate the product sigma algebra. The same statement holds after completion because completed-measurable L^2 functions have product-measurable representatives. Hence $P_\alpha F = \mathbb{E}[F \mid \mathcal{A}_\alpha]$ converges in L^2. Conditional expectation is contractive in L^∞ and L^2. For a coordinate included in $\mathcal{A}_\alpha$, its averaging projection commutes with P_α because all operations integrate independent coordinates; for an excluded coordinate $D_i P_\alpha = 0$. Consequently $\mathrm{I}_i(P_\alpha F) \le \mathrm{I}_i(F)$ in either case. Applying P_α to a Hoeffding component supported on S either annihilates it (if an active coordinate is excluded) or preserves the same support with the single-coordinate functions replaced by conditional averages. Thus it preserves the closed degree-d subspace. Finally, expand each finite projected function into its Hoeffding components and sum the influences over any finite coordinate set. Passing to the L^2 limit yields the final inequality. □

Completion of Theorems 2.2 and 2.9. Apply the finite theorem to every bounded finite-partition projection; the projected variance tends to the true variance and the maximum influence is at most $M = \sup_{i \in J} \mathrm{I}_i(F)$. Passage to the limit proves the quantitative inequality with M in place of the maximum. If $V(F) > 0$, the lower bound implies $M > 0$. The set $\{i : \mathrm{I}_i(F) \ge M/2\}$ is finite: if it contained arbitrarily many indices, the finite-subset influence bound would exceed $2V(F)$. Every coordinate outside this finite set has influence less than $M/2$, so the supremum is attained inside the

set. If $V(F) = 0$ the inequality is immediate. The rank-r case follows after identifying its effective output subspace with $\mathbb{R}^r$. □

Assumptions that cannot be dropped. Product independence is essential to the independent fourth-moment step. It is also essential to the posterior preservation of the unobserved law used in Section 6. A correlated law can have a large variance but vanish under single-coordinate conditional resampling when the other coordinates nearly determine the coordinate. In particular, the theorem is *not* asserted for arbitrary correlated inputs. The degree-two condition also cannot be replaced by unrestricted degree without proving an additional independent influence theorem; we do not import the all-degree Turturean route.

12 Prior art, original contribution, and evidence level

12.1 Scope-matched prior work

(i) **Hoeffding decomposition on arbitrary products.** The decomposition, noise projection, and resampling identities are established background. In particular Blais–O'Donnell–Wimmer [4] explicitly develop orthogonal expansion techniques for arbitrary categorical product distributions. The present claim is quantitative, not the existence of ANOVA itself.

(ii) **Decision-tree influence.** O'Donnell–Saks–Schramm–Servedio [3] prove an inequality for decision trees and discuss arbitrary product input and non-Boolean output extensions. We do not claim the observation drifts or influence comparisons as a first use of decision-tree methods, nor an efficient implementation beyond the stated existence.

(iii) **Uniform Boolean cube and distributional variants.** The conjecture of Aaronson–Ambainis [5] was resolved in its all-degree Boolean-cube form by Turturean [11]. Girish–Mittal–Nehoran–Raz [7] prove equivalences with fixed-bias Boolean products and slices. These results are not used as black boxes here. A fixed-bias equivalence alone does not assert the identical numerical constant for arbitrarily small biases and arbitrary alphabet cardinalities.

(iv) **Quadratic Grothendieck and completely bounded norms.** Classical quadratic Boolean inequalities can give substantially better constants in their own settings; Grothendieck principles and completely bounded polynomial estimates [12, 13] operate in different input/norm regimes. A direct categorical tensor norm implication, or an earlier bounded-backdoor theorem with the same hypotheses, has not been excluded by this limited prior-art search; the novelty of the sharp-scope quadratic inequality is therefore not treated as fully adjudicated.

12.2 What is structurally new in the proposed synthesis

The primary proposed mechanism is a matching certificate for quadratic Hoeffding interactions, followed by a *two-level* backdoor principle. A hitting set for all high-order supports gives an influence transfer, while the stronger bound on the number of unobserved coordinates in each support permits a direct adaptive observation reduction. The distinction is mathematically essential: Proposition 4.4 gives a parity family that separates the two statements. Independent cubic blocks add a separate range-budget argument, producing a uniform constant even when the union of all cubic covers is

large. Claims of global originality for these principles await historical review; they are not asserted as first discoveries merely because the present proofs are independent.

12.3 What was proved without importing Turturean

The independent quadratic proof has four mathematical steps: (1) the disjoint-matching even/odd coupling, (2) dimension-free boundedness budgets obtained from quadratic interpolation and fiber perturbations, (3) a vertex assignment plus Hilbert fourth-moment inequality, and (4) an exact weighted posterior energy identity followed by a rank-compressed maximal influence inequality. Each step appears in this article with proof; the finite-to-general passage is included. No theorem from [11], no source code from his Lean proof, and no auxiliary cube certificate inequality enters any derivation.

12.4 Evidence status and necessary independent attack

The theorem proofs above, including the higher-degree backdoor transfers, are analytic. An exact-arithmetic checker shipped with this draft verifies a finite collection of independent falsification examples: input masses, variance, influences, matching budgets, and posterior drift. Such tests are evidence against elementary mistakes in those instances, not certificates of universal claims. The finished manuscript has *not* been checked by the Lean kernel and has not been reviewed by an independent specialist.

The highest-value next audit would independently rederive the matching budget with arbitrary categorical alphabets, check that every unretained edge admits the required endpoint assignment, verify the $3s_i^2$ Hilbert fourth-moment coefficient, and test the posterior weighted drift on mixed first- and second-order examples. A separate prior-art audit should examine categorical bilinear operator norms and vector-valued Rademacher/Grothendieck inequalities before any worldwide-first claim. If an older theorem implies the quadratic bound without changed hypotheses, the appropriate novelty classification must be lowered.

13 Conclusion and the remaining frontier

There are two complementary structural operators in this theory. The first is the *matching–fourth-moment certificate*: bounded quadratic interactions on arbitrary independent product laws force a coordinate with variance-squared influence, with no dependence on atom masses or alphabet dimensions. Its rank-sensitive counterpart and exact posterior energy decrement yield sharp-order whole-symbol query budgets and spectrum-adaptive recovery.

The second is the *quadratic backdoor*. Averaging a small transversal of the high-order supports proves a variance-squared influence theorem without any upper bound on the original interaction degree. Actually observing a stronger sectional backdoor reduces the conditional target to quadratic degree and gives a rank-sensitive recovery budget with only an additive backdoor overhead. At degree three the two conditions coincide. Independent covered blocks permit an unbounded number of separate cubic networks without introducing an alphabet factor. The parity and address examples explain why the conditioning hypothesis and adaptive access model cannot be dropped.

Neither these results nor the known all-degree Boolean-cube influence theorem proves exponent two for arbitrary cubic support hypergraphs. The next decisive mathematical test is a uniform assignment or overlap certificate for interlocking cubic links. The remaining scientific tasks are a scope-matched categorical Grothendieck prior-art audit, independent review of the global matching proof and the posterior transfer, and end-to-end formal verification. No step of the present argument imports Turturean's theorem or his proof machinery.

A A Hermitian tensor inequality for binary quantum contrasts

We supply the simple-tensor step used only for the binary-source refinement of Proposition 10.3. Let b_i be faithful density matrices and H_i arbitrary Hermitian matrices. Define $Q_b(H) = \mathrm{Tr}(H J_b^{-1}(H))$. Then

$$Q_{b_1 \otimes b_2}(H_1 \otimes H_2) \geq Q_{b_1}(H_1) Q_{b_2}(H_2). \tag{34}$$

Proof. Diagonalize b_1, b_2. Expand in ordered matrix-unit entries $H_i[p, q]$. For a Hermitian H_i, the squared magnitude $|H_i[p, q]|^2$ is unchanged by reversing the order of p and q. Give each unordered pair of indices probability proportional to its squared matrix-entry mass. Conditional on those unordered pairs, independently reverse each ordered pair with probability $1/2$. The global inverse-Sylvester coefficient is $2/(\lambda_{1,p}\lambda_{2,u} + \lambda_{1,q}\lambda_{2,v})$. Its expected denominator under independent reversals is $\frac{1}{2}(\lambda_{1,p} + \lambda_{1,q})(\lambda_{2,u} + \lambda_{2,v})$. Jensen's inequality for the convex function $x \mapsto 1/x$ implies that the averaged global coefficient is at least $4/[(\lambda_{1,p} + \lambda_{1,q})(\lambda_{2,u} + \lambda_{2,v})]$, which is the product of the two local inverse-Sylvester coefficients. Summing with the nonnegative squared-entry weights gives (34). The argument is an orientation average, not a claim that the Sylvester map tensorizes as an operator identity. □

Every centered classical contrast space for a nontrivial binary site has real dimension one. Therefore a pure pair interaction is a scalar multiple of a simple tensor of centered contrasts. Its encoded signal is a Hermitian simple tensor, so the inequality applies. For three-symbol or larger sites, a generic interaction is a *sum* of simple tensors and the stronger operator inequality does not follow; this is exactly the source of the possible extra transfer penalty.

B Independent finite-model checker protocol

The bundled scripts `check_quadratic_models.py` and `check_structured_backdoors.py` uses rational probability masses for small heterogeneous product spaces, explicit Hoeffding component tables, and exact fraction arithmetic wherever possible. It tests the variance identity, each resampling influence, and the one-step posterior degree-weighted potential identity. They also check exact backdoor energy splits, cubic address influences and conditional quadratic sections. The quadratic checker enumerates matchings for the interaction sup-norm bound and checks output sharpness and addressing examples. Any success flag is expressly scoped to these finite instances. A failed assertion is a mathematical counterexample candidate or an implementation defect requiring investigation before use of the associated theorem. This paper makes no claim of a universal machine certificate.

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